Forecasting Product Demand Using Multivariate Time Series Models

Presented By : Shruti Patil

Content

Background & Motivation

- Retailers rely on daily sales forecasts to plan inventory and staffing, and poor forecasts lead to stock issues and inefficiencies

- Store-level forecasting needed: locations vary in size, promotions & patterns

- Personal motivation: hands-on learning of time series forecasting for data science careers

Problem Statement

How do training window selection, model stability over time, and training data length shape forecast accuracy for an individual store & how do results vary when the same fixed setup is applied across all stores?

Research Questions

1. What is the best training window for one store for fixed test window?

2. Is the model structure remain stable over time for same store?

3. What's the impact of training data length for one store?

4. How do results vary across different stores under a fixed training and test setup

Data

- Rossmann Store Sales Kaggle competition — a real German drugstore chain
- 1 January 2013 to 31 July 2015 (~ 2.7years)
- 1,115 stores
- 18 variables total after merging 2 CSV's
- ~ 1.02 million daily observations

Train.csv

Transactional data

- Daily sales data for each store
- Target variable: Sales
- Store status (Open / Closed)
- Promotion activity (Promo)
- Holiday information (State & School)
- Time features (Date, Day of Week)

Store.csv

Static store characteristics

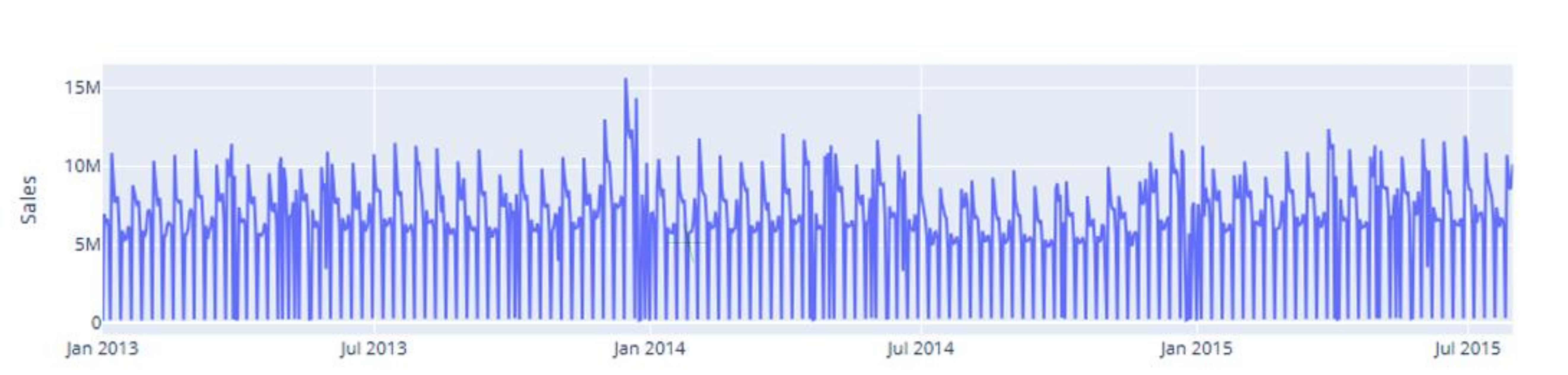
- Store type and assortment level
- Competition distance
- Competitor opening information
- Continuous promotion program (Promo2)
- Promotion timing (PromoInterval)

Data Quality assesment

- Core variables — Sales, Open, Promo, Customers — 100% complete
- Missingness in data:
  - Promo2 - 50% — stores simply did not join the continuous promotion programme
  - Competition date: -31.8% of stores — competitor opening dates unavailable
- Store operational Status : 15.9% of obseravtions(days)show zero sales — stores were closed

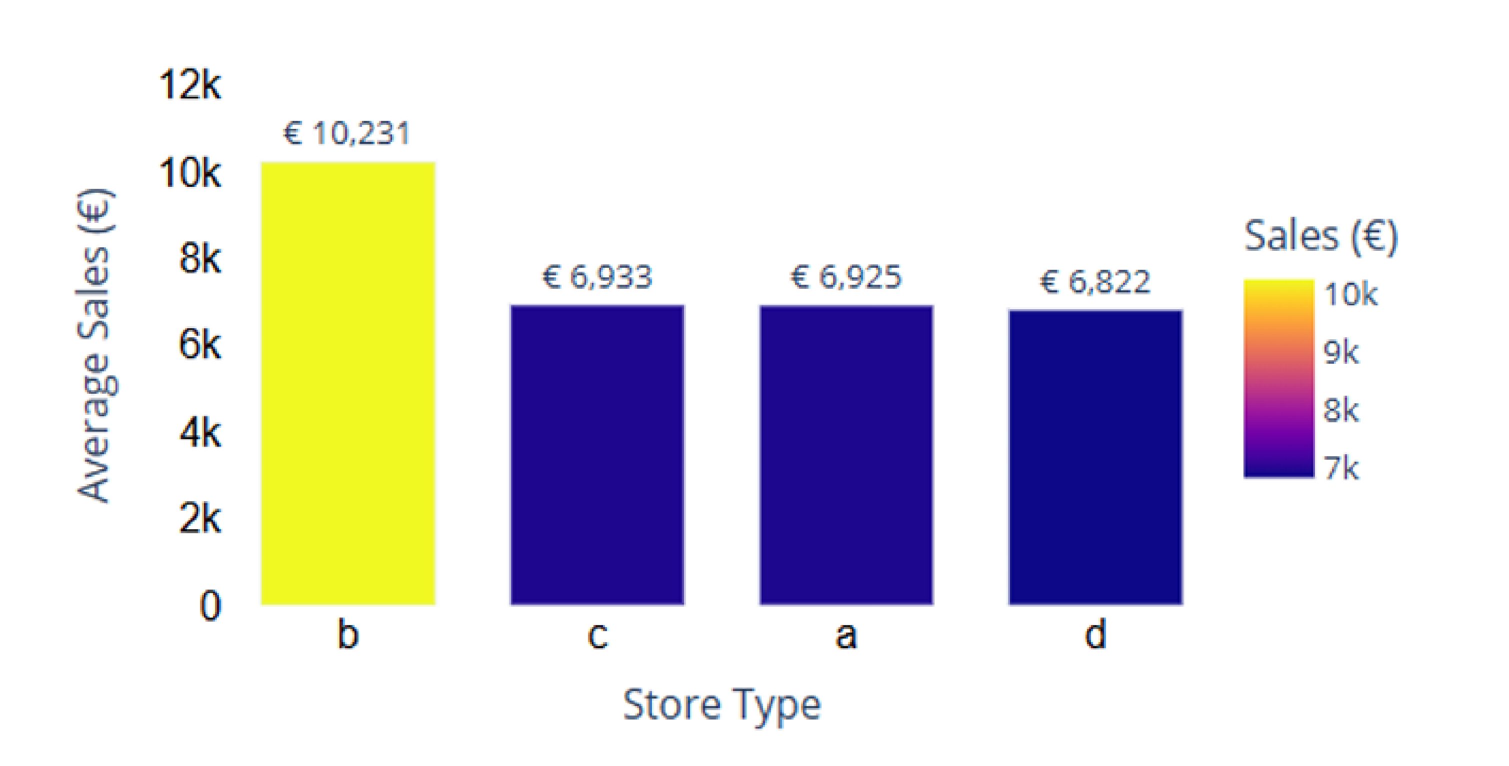
EDA

Global Sales Trends



- Stable sales level across 2013–2015 with strong recurring weekly cycle
- Sharp drops on Sundays & public holidays · spikes around year-end

Average sales by store type

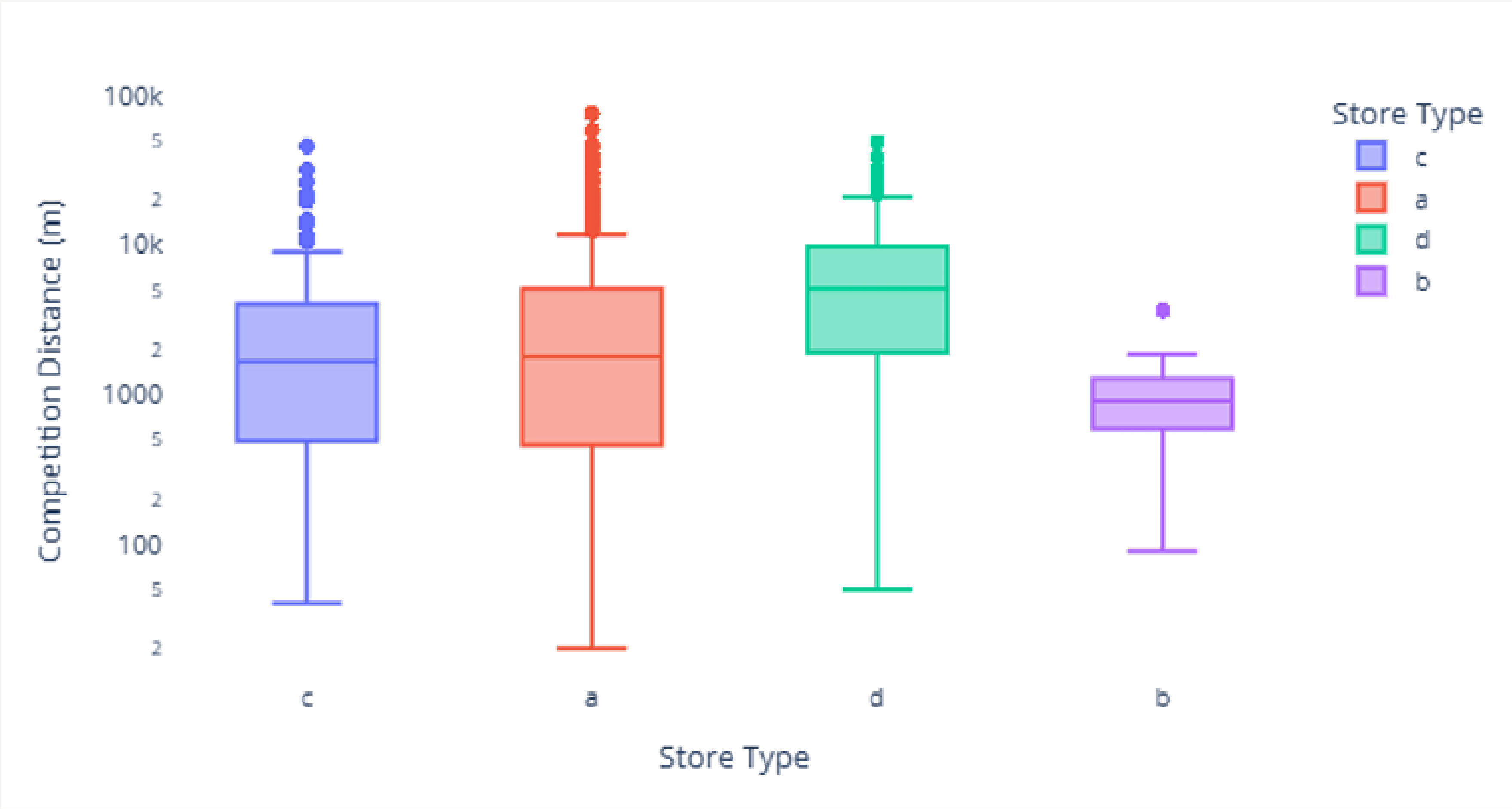


- Store Type B dominates · €10,231 avg daily sales · nearly 50% higher than others
- Store Types A, C, D similar · €6,800 to €6,900 range

Sales vs Customer by Store Type

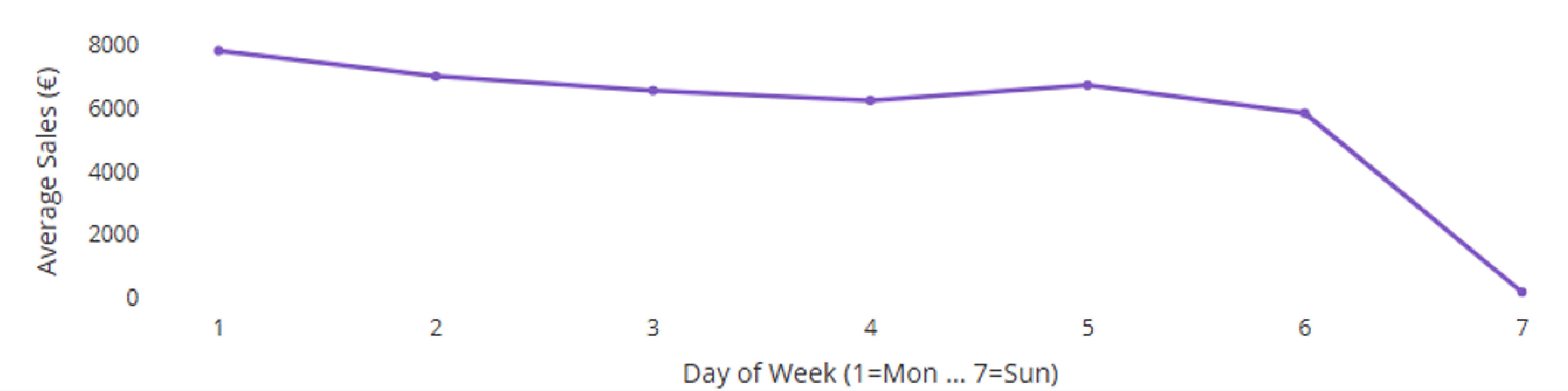
- More customers = higher sales across all store types — strong positive relationship
- Store Type B reaches almost 6,000 customers per day — highest traffic and revenue

Competition distance by store type



- Store Type D located furthest from competitors — least competitive pressure
- Store Type B has shortest competition distances — operates in highly competitive areas

Average Sales by Day of the Week



- Monday & Friday, Saturday - highest average sales days
- Sunday - near-zero sales ; most stores closed

Average Sales by Month and Year

- December · highest sales · clear year-end seasonal spike
- Jan–Feb · lowest sales · consistent across all three years

Feature Engineering

- Handling Missing Values
  - Competition Distance: 2,642 missing → filled with max distance observed (assumed far from competitors)
  - Competition opening dates: missing → set to 0 (unavailable)
  - Promo2: missing → set to 0 (store not in promotion programme)
- Encoding
  - Store Type & Assortment → Categorical features one-hot encoded to avoid multicollinearity
- New Features
  - IsPromoMonth: binary flag (1 = promotion month, 0 = not)
  - Calendar features extracted from Date: Year, Month, DayOfWeek

Classical Time Series Diagnostics (Store 1)

- Decomposition (STL, m = 7)
  - Stable weekly seasonality, no strong long-term trend
- Stationarity (ADF Test)
  - ADF = −4.37, p = 0.0003 → Series is stationary (d = 0)
- ACF & PACF
  - Spikes at lags 7, 14, 21, 28 → m = 7
  - PACF shows strong spike at lag 1 → previous day most important

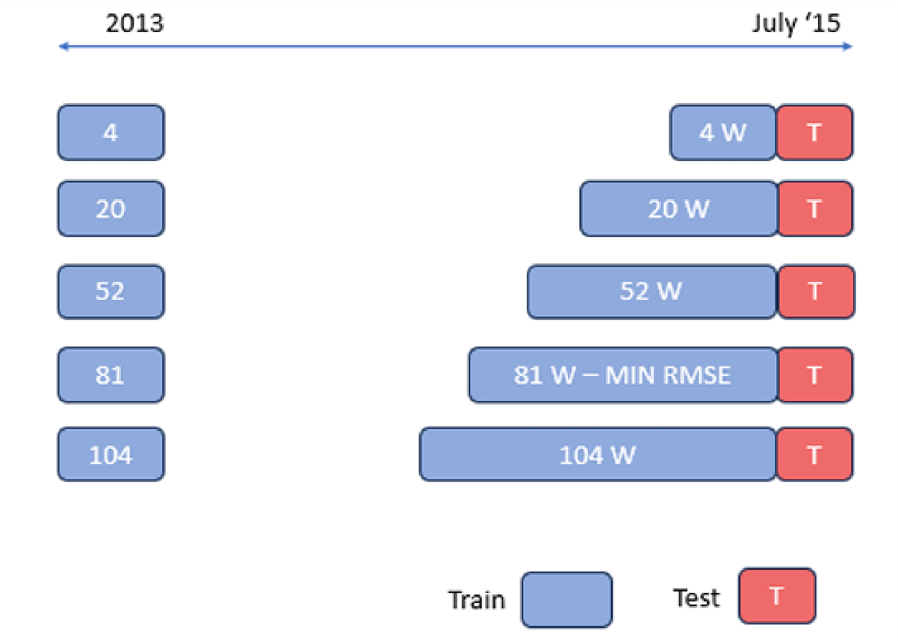
Methodology

Stage 1

Selecting best training window for store 1 on fixed test window.

Method

Expanding window search

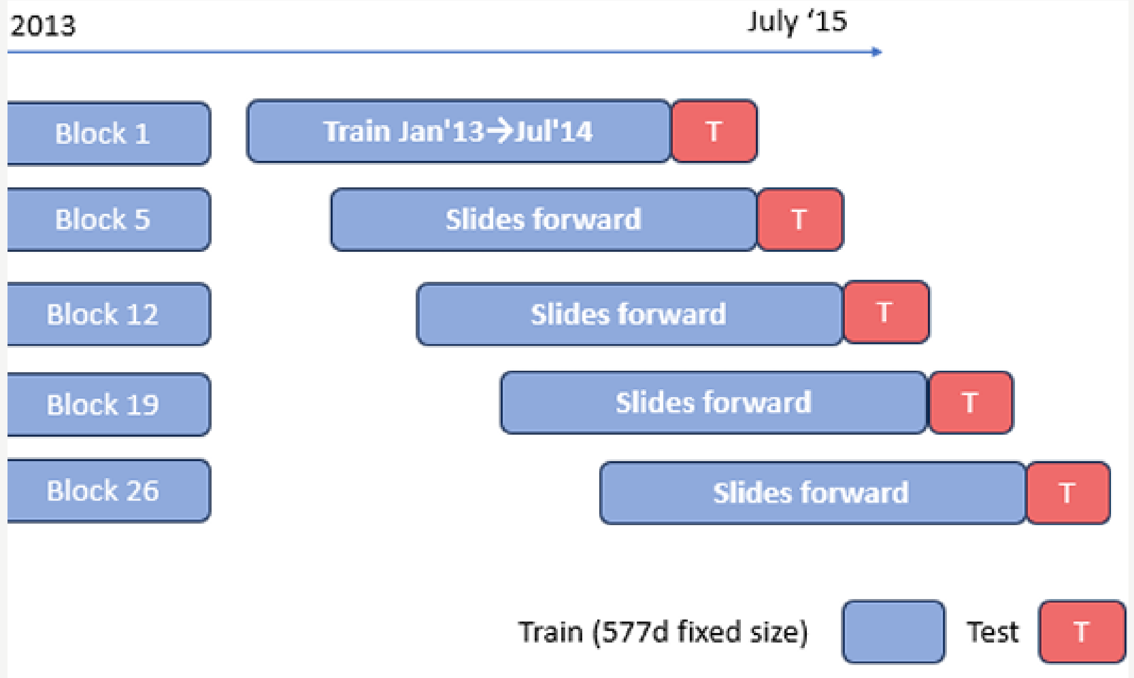


Stage 2

Model structure remains stable over time for store 1

Method

Forward sliding window

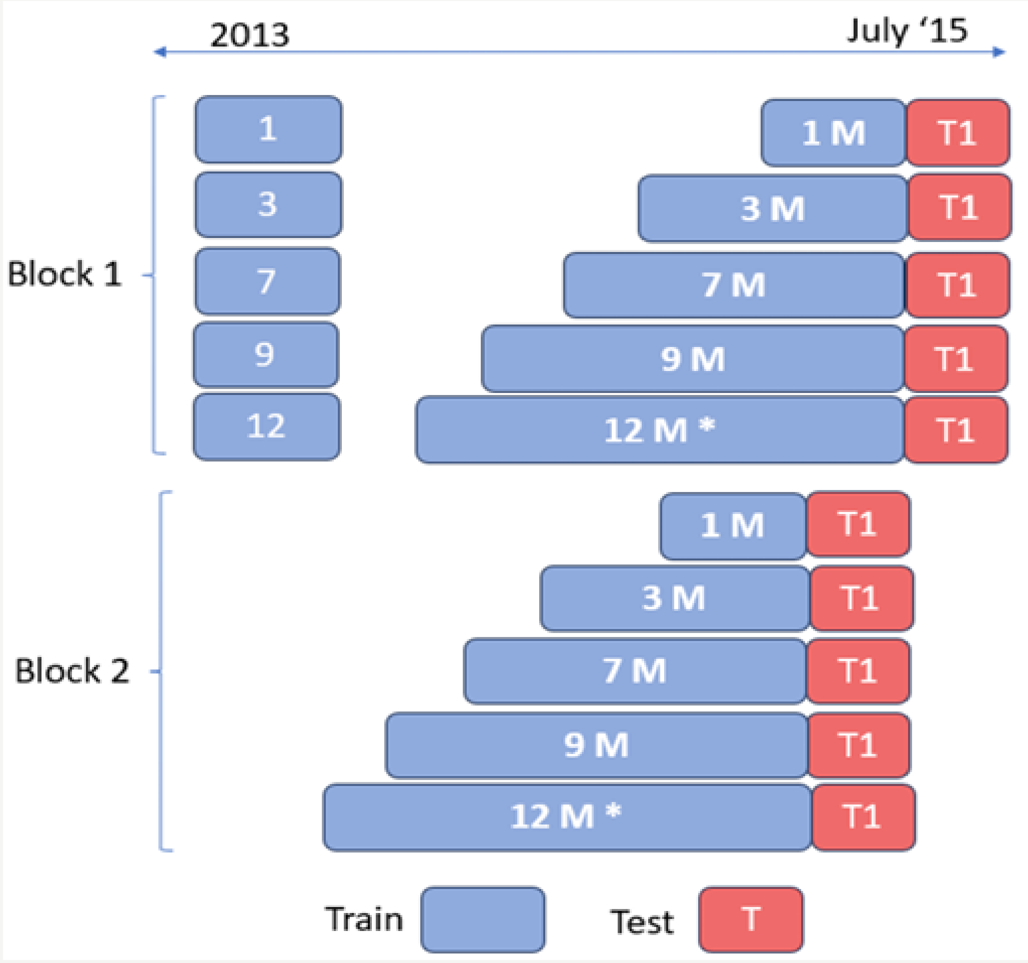


Stage 3

Influence of Training Data Length for Store 1

Method

Backward evaluation blocks



Stage 4

Forecasting results across stores

Method

Fixed training and test configuration

Fixed Training Window & Test Period

Implementation

Tools & Libraries

- Language · Python
- SARIMAX model · statsmodels library
- AutoARIMA order selection · pmdarima library
- Data manipulation · pandas and NumPy
- Performance metrics · scikit-learn

Key Parameters — Same Across All Stages

- Seasonal period $m = 7$ · captures weekly cycle
- Maximum $p, q = 3$
- Maximum $P, Q = 2$
- Non-seasonal differencing $d = 0$; series already stationary
- Seasonal differencing $D = 0$ or 1
- Maximum 20 models evaluated per window
- Significance threshold $p \le 0.05$ for variable selection
- Model selection criterion · AIC

Implementation — Stage by Stage Execution

Stage 1 · Training Window Selection · Store 1

- Evaluated 101 training windows · 4 to 104 weeks ; one week increments
- AutoARIMA + backward elimination per window
- Forecasts generated for fixed 14 day test window
- Metrics · RMSE
- Window with lowest RMSE selected

Stage 2 · Model Stability · Store 1

- 26 forward sliding blocks · training fixed at 577 days · test window 14 days
- Both windows shift forward by 14 days per block
- AutoARIMA + backward elimination run per block
- Most frequent model configuration across 26 blocks = stable model

Stage 3 · Training Length · Store 1

- Model structure fixed from Stage 2 · only training length varied
- 26 backward blocks · 1 to 12 months training length evaluated per block
- Mean RMSE calculated per training length

Stage 4 · Network-Wide Forecasting · All 1,115 Stores

- Same pipeline (AutoARIMA + backward elimination) applied to all 1,115 stores
- Fixed: 81-week training window · 14-day test period
- Results compiled into structured dataset for network-wide analysis

Results for stage 1

Selected Model: SARIMAX(0,0,0)(2,0,0)[7] · Optimal Training Window: 81 weeks

Actual VS Predicted Sales

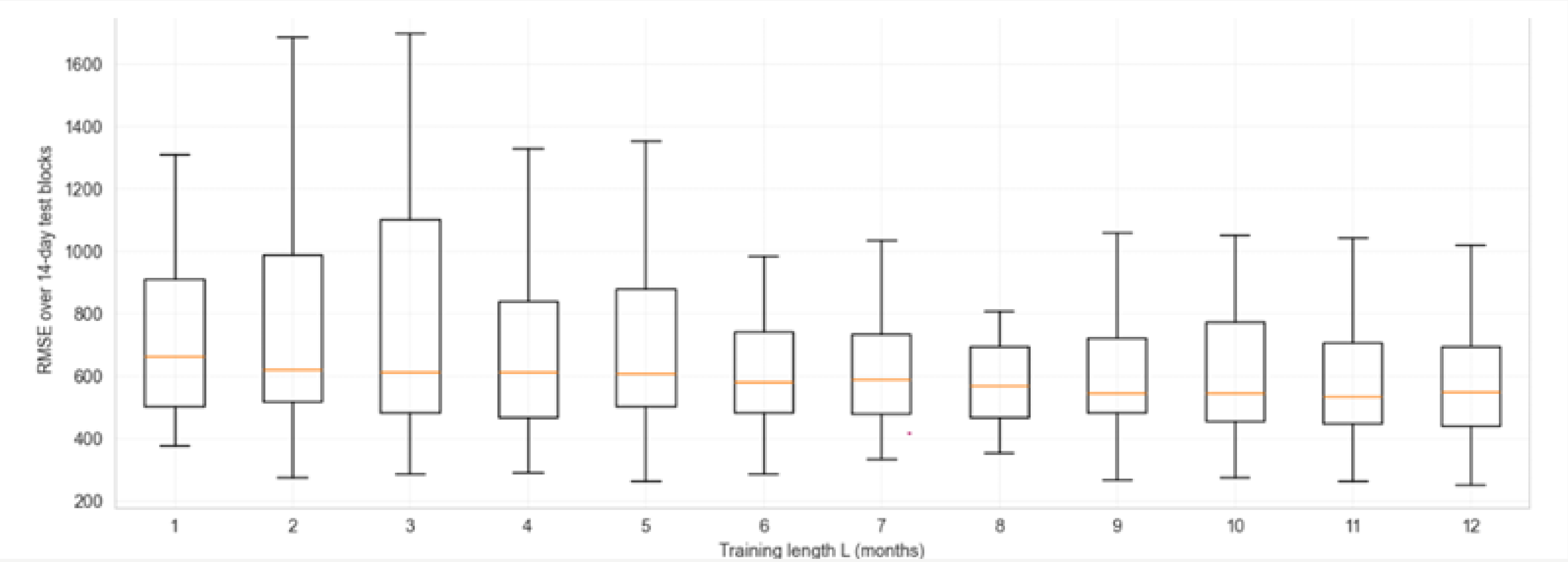
- Metrics
  - MAPE 4.7% — forecast errors relatively small
  - R^2 0.983 — model explains 98.3% of sales variation
  - RMSE €225
  - MAE €179 — mean absolute error
- Exogenous Variables Retained
  - Open · Promo · DOW_2 · DOW_3 · DOW_4 · DOW_5 · DOW_6 · SchoolHoliday
- Key Takeaways
  - Sundays correctly predicted as 0 (store closed)

Results for stage 2

Rank	SARIMAX Order	Frequency	Avg RMSE
1	(0,0,1) (2,0,2,7)	4/26	485.95
2	(0,0,0) (2,0,0,7)	3/26	950.18
3	(0,0,3) (2,0,2,7)	2/26	601.69

- Exogenous Variable Selection
  - Open, Promo, DOW_2–DOW_6 selected in all 26 blocks (100%)
  - DOW_7 never selected — Open already handles Sunday closure
  - SchoolHoliday (4/26), IsPromoMonth (3/26), StateHoliday (1/26) — too irregular
- Most Stable Model
  - SARIMAX(0,0,1)(2,0,2)[7]

Results for stage 3



	Mean_RMSE	Std_RMSE
1	831.81	544.26
2	816.40	495.12
3	803.02	452.66
4	757.21	443.43
5	759.63	412.82
6	744.08	460.05
7	731.81	422.04
8	733.92	448.99
9	720.42	436.53
10	715.49	420.45
11	708.96	422.48
12	703.01	426.57

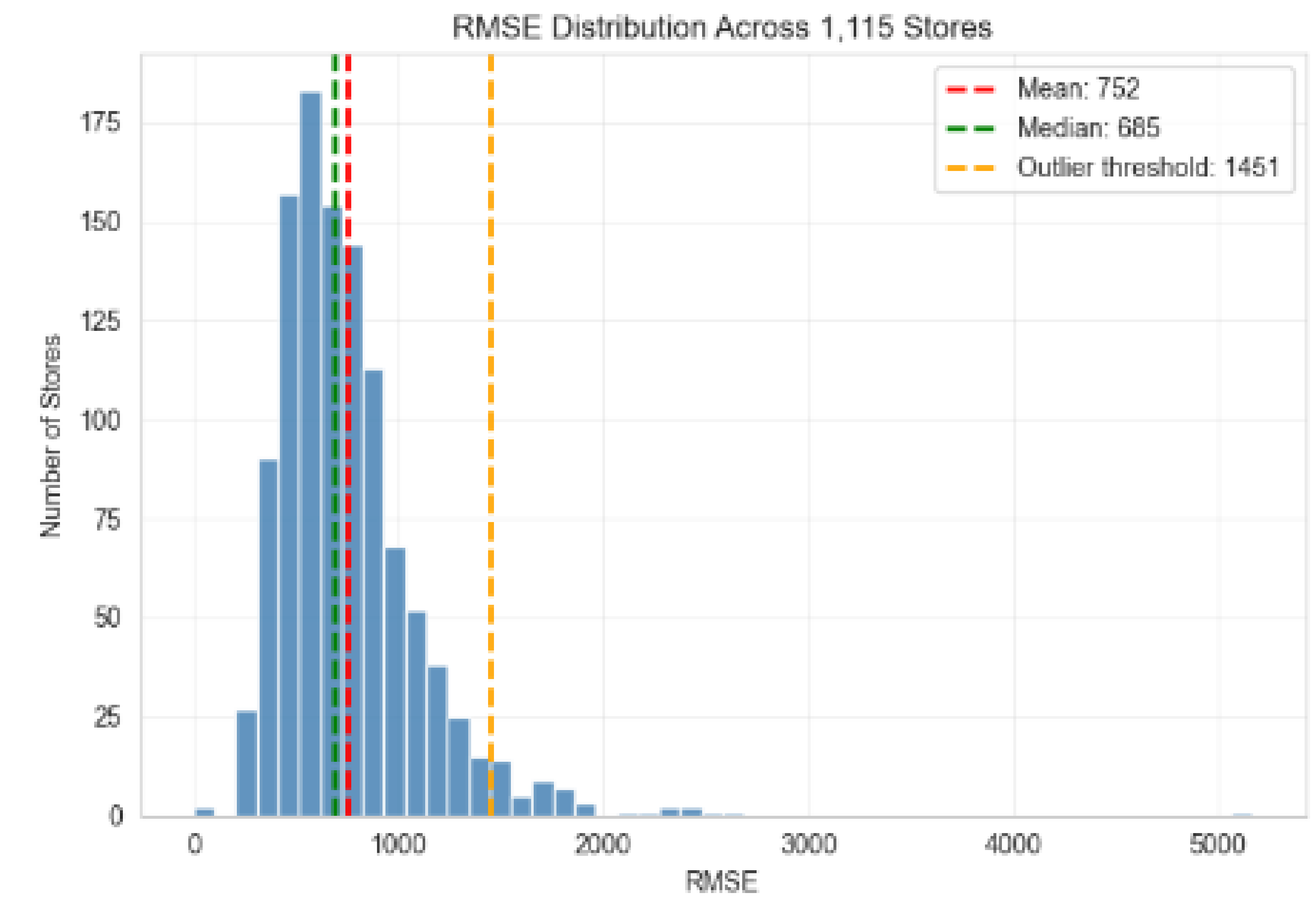
- RMSE steadily decreases from month 1 to 6
- After 6 months · boxplots show similar medians and spread
- Month 8 (733.92) slightly higher than month 7 (731.81) — more data does not always mean better
- Variability stays similar from month 7 onwards · Std around 420–450
- RMSE of 250.83 on final 14-day test · comparable to Stage 1 benchmark of 225.35

Results for stage 4

Overall Performance

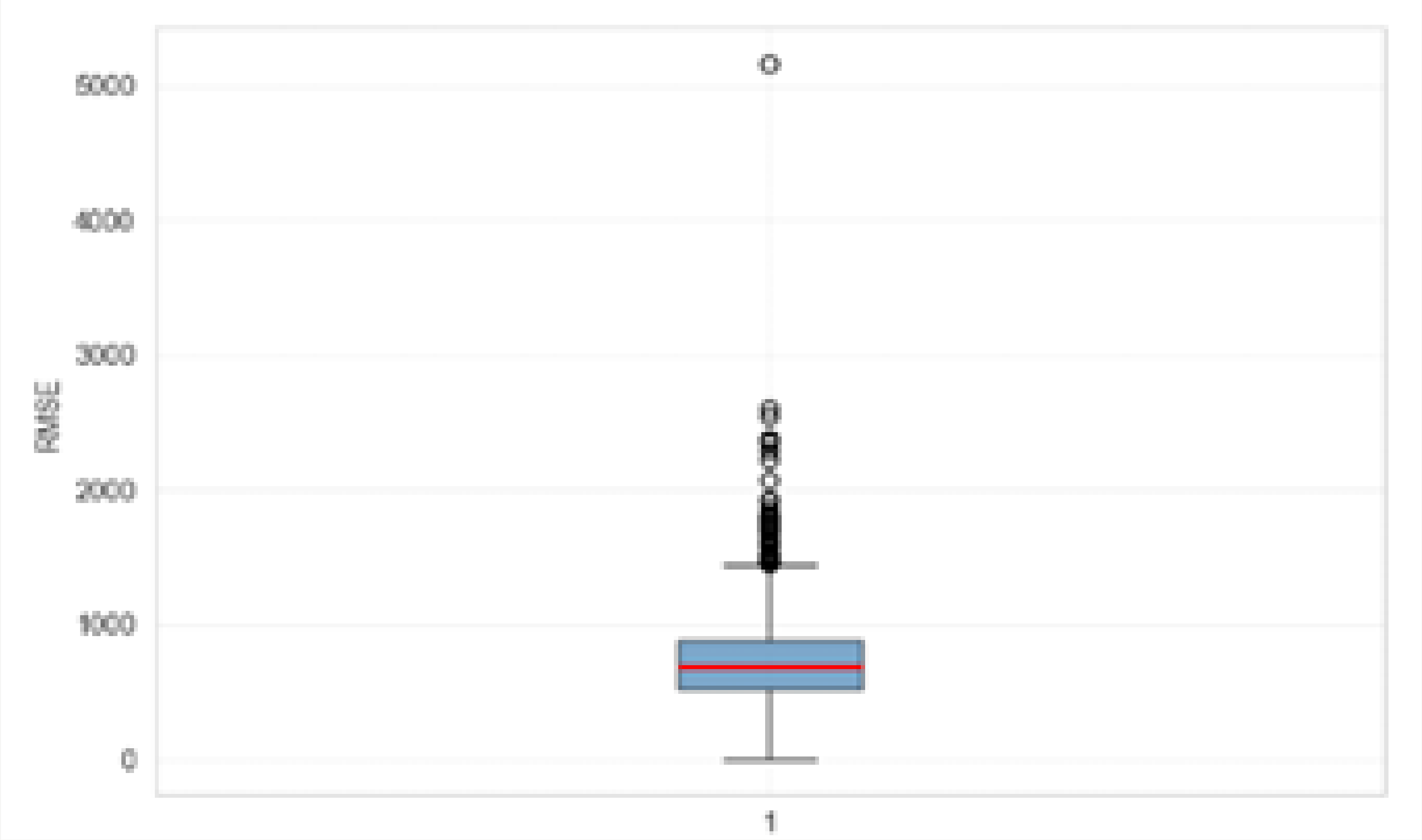
- Mean RMSE · 751.55 · Median RMSE · 684.62 · Standard deviation 359.75 · Maximum RMSE · 5164.97
- Median is lower than mean — right skewed distribution · few high error stores pull average up

RMSE Distribution



- Majority of stores cluster between RMSE 500 and 900
- Small subset of outliers drives the right skew · outlier threshold = 1451

Outlier Analysis



- 47 stores (4.2%) exceed RMSE threshold of 1450.60
- 43 of 47 outlier stores (91.5%) belong to Always Closed on Sunday category

Store Type Analysis

Type	Count	Avg RMSE	Avg MAPE	Avg Sales
a	602	754.14	10.19	5753.09
b	17	906.66	6.98	10424.88
c	148	774.96	9.88	5744.61
d	348	729.53	9.00	5746.94

- Store Type B · highest RMSE (906) but lowest MAPE (6.98%) — due to higher sales volume
- Store Type D : most stable · lowest RMSE (729) · MAPE 9.00%
- Store Types A and C - similar mid range performance · RMSE ~754–775

Sunday Trading Patterns

Sunday Category	Count	Avg RMSE
Always Closed	1082	743.65
Always Open	10	1017.61
Sometimes Open	23	1007.14

Frequency of Exogenous Variable Selection

```
==========================================
EXOGENOUS VARIABLE IMPORTANCE
==========================================
     Variable  Count  Pct_Stores
         Open   1115       100.0
        Promo   1115       100.0
        DOW_4   1083        97.1
        DOW_3   1081        97.0
        DOW_2   1061        95.2
        DOW_5   1010        90.6
        DOW_6    913        81.9
SchoolHoliday    600        53.8
 StateHoliday    321        28.8
 IsPromoMonth    182        16.3
        DOW_7     39         3.5
```

Model Architecture

Rank	Model Type	Count	Avg RMSE	Avg MAPE	%
1	(0,0,1)(2,0,2,7)	169	705.00	9.33	15.2%
2	(0,0,0)(2,0,2,7)	87	800.18	9.72	7.8%
3	(0,0,1)(2,0,0,7)	78	644.44	10.19	7.0%
4	(0,0,1)(2,0,1,7)	60	810.85	8.74	5.4%
5	(0,0,3)(2,0,2,7)	55	839.56	11.09	4.9%
6	(0,0,1)(0,0,2,7)	51	722.34	8.59	4.6%
7	(0,0,3)(2,0,1,7)	41	805.12	10.90	3.7%
8	(0,0,0)(2,0,0,7)	39	602.21	10.59	3.5%
9	(1,0,0)(2,0,1,7)	38	599.72	9.23	3.4%
10	(1,0,1)(2,0,1,7)	38	756.91	8.48	3.4%

- Most common model - SARIMAX(0,0,1)(2,0,2)[7]
- Used in only 169 stores (15.2%)
- No single model fits all stores
- Each store has its own unique sales dynamics

Conclusion

- SARIMAX with automated selection delivers reliable short-term forecasts
- Weekly seasonality ($P=2$, $m=7$) dominant in 91.7% of stores · Open & Promo universal predictors
- Training data beyond a certain point does not significantly improve performance
- Each store behaves differently, automated store-level modeling is necessary to achieve good forecasting accuracy

Future Work

- Add external variables · regional events
- Explore ML models · hybrid SARIMAX + ML approaches
- Test on longer forecast horizons beyond 14 days
- Validate framework on other retail chains beyond Rossmann

Thank You!